Sébastien Kleff

Postdoctoral researcher at Inria

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Summary

I am a roboticist working at the intersection of robotics control, modeling, numerical optimization and learning. My long-term vision aims at endowing robots with an advanced sense of touch so that they can effectively benefit humans in the real world (e.g., product recycling, assistance to manual labor). My research builds on 8+ years of international experience in top laboratories across France, USA, Germany, and China.

Research experience

Postdoctoral researcher — **Bordeaux, France**
Inria — *Since Sep. 2024*
Robotics research at AUCTUS team.

Postdoctoral associate — **NY, USA**
New York University (NYU), Electrical & Computer Engineering (ECE) Dep. — *May 2024 – Aug. 2024*
Robotics research at Machines in Motion Laboratory.

Research assistant — **NY, USA & Toulouse, France**
NYU & LAAS-CNRS — *Sep. 2019 – Apr. 2024*

- NYU ECE. Doctoral research at Machines in Motion Laboratory (Sep. 2019 – Oct. 2020, then Sep. 2022 – Apr. 2024).
- LAAS-CNRS. Doctoral research at the Gepetto team (Oct. 2020 – Aug. 2022). Two-year international PhD mobility in the context of a collaboration with NYU. I was employed by CNRS and carried out part of my doctoral research on-site.

Research engineer — **Compiègne, France**
Heudiasyc – UTC/CNRS — *Mar. 2019 – Aug. 2019*
Research engineer at SyRI team in the EU project ESCAPE, in partnership with Renault. Developed and deployed a map-matching algorithm for autonomous vehicle localization, validated on Renault ZOE experimental vehicles.

Research intern — **Tübingen, Germany**
Max Planck Institute for Intelligent Systems, Autonomous Motion Department — *Sep. 2015 – Feb. 2016*
Movement Generation and Control Group. Supervised by Alexander Herzog. Research on optimal control for bipedal locomotion, which led to a peer-reviewed publication with Majid Khadiv [C9].

Education

PhD in Electrical Engineering — **New York, USA**
New York University, Electrical & Computer Engineering Dep. — *Sep. 2019 – May 2024*
Thesis: *Toward Force Feedback in Model-Predictive Control*. PhD awarded on 15 May 2024.
Doctoral advisor: Ludovic Righetti (NYU). Scientific advisor during international mobility: Nicolas Mansard (LAAS-CNRS).

Master of Science & *Diplôme d'Ingénieur* — **Nantes, France & Shanghai, China**
École Centrale de Nantes (ECN) & Shanghai Jiaotong University (SJTU) — *Sep. 2013 – Apr. 2019*

- ***Diplôme d'Ingénieur***, ECN (Sep. 2013 – Apr. 2019). General engineering curriculum with specialization in robotics.
- MSc in Control Science and Engineering, SJTU (Sep. 2016 – Mar. 2019). Long-term international mobility in the context of a double-degree program between ECN and SJTU. Advanced coursework in control science and master's research on reachability analysis under the supervision of Prof. Ning Li, which led to 2 peer-reviewed publications [J4], [C8].

Bachelor of Science — **Nantes, France & Compiègne, France**
École Centrale de Nantes & Lycée Pierre d'Ailly (LPA) — *Sep. 2011 – Aug. 2013*

- BSc in Engineering. ECN (Sep. 2013 – Apr. 2014). General science and engineering curriculum.
- *Classes Préparatoires*. LPA (Sep. 2011 – Aug. 2013). Two-year intensive undergraduate program in mathematics and physics, preparing for the national competitive entrance examination to the French *Grandes Écoles d'Ingénieurs*.

Awards and scholarships

Seal of Excellence
Marie-Skłodowska-Curie Action Postdoctoral Fellowship *Feb. 2025*

Best poster award
Stagewise Implementations of Sequential Quadratic Programming for MPC [J2] *May 2024*
2nd Advancements in Trajectory Optimization and Model Predictive Control for Legged Systems Workshop at ICRA

China Scholarship Council (CSC) for international students
Funder: China's Ministry of Education. *Sep. 2016 – Feb. 2019*
Amount: approx. 3000 RMB/month. Duration: 30 months. Acceptance: 10-30%.

"Envoléo" scholarship
Funder: Région Pays de la Loire. *2015*
Amount: approx. 1k EUR. For student international mobility.

Selected scientific contributions

Nonlinear Model-Predictive Control (NMPC) for Robotics: I contributed to the democratization of NMPC in robotics through theoretical and experimental contributions. I developed an efficient numerical solver and led its open-source release [J2]. The library is now a standard back-end of the `crocoddyl` ecosystem, used by 20+ researchers across several labs, and is being integrated into `acados`. I also proposed the first NMPC running at 1 kHz on a torque-controlled manipulator [C7] (∼123 citations) and addressed key NMPC limitations – short planning horizon, sensitivity to local minima – using reinforcement learning [J3], [C5].

Unified framework for NMPC and Force Control: I made theoretical and experimental contributions to NMPC for contact-rich tasks, identifying a long-standing structural limitation of standard NMPC for force regulation and proposing complementary formulations [C3], [C2], [C4] restoring feedback from force measurements inside the predictive loop. The framework unifies motion anticipation, constraint handling, and contact-force regulation [JR1], and was validated industrially with Airbus and PAL Robotics in the EU project AGIMUS [C1].

Structured Tactile Control: I initiated a new research axis at Inria around structured tactile control. I leveraged high-resolution tactile sensing and model-based control through physically-grounded tactile features. I developed a complete pipeline and showed that this method can regulate physical interaction up to 5× more accurately than existing methods [J1], which opens a concrete path toward safe and robust real-world manipulation.

Publications

Journal articles

[J1]: **S. Kleff**, L. Joseph, V. Padois. "Geometrically Consistent Tactile Servoing via Hybrid Force-Position Control at the Center of Pressure". *IEEE Robotics and Automation Letters*, vol. 11, pp. 8928–8935, 2026.

[J2]: A. Jordana*, **S. Kleff***, A. Meduri*, J. Carpentier, N. Mansard, L. Righetti. "Structure-Exploiting Sequential Quadratic Programming for Model-Predictive Control". *IEEE Transactions on Robotics*, vol. 41, pp. 4960–4974, 2025.

[J3]: A. Jordana, **S. Kleff**, A. Haffemayer, J. Ortiz-Haro, J. Carpentier, N. Mansard, L. Righetti. "Infinite-Horizon Value Function Approximation for Model Predictive Control". *IEEE Robotics and Automation Letters*, vol. 10, pp. 7563–7570, 2025.

[J4]: **S. Kleff**, N. Li. "Robust Motion Planning in Dynamic Environments Based on Sampled-Data Hamilton-Jacobi Reachability". *Robotica*, vol. 38, 2020.

Journal articles under review

[JR1]: **S. Kleff**, A. Jordana, R. Khorrambakht, N. Mansard, L. Righetti. "Force Feedback Model Predictive Control: A Soft Contact Approach". Submitted to *Autonomous robots*, 2026.

The * symbol denotes equal contribution.

Peer-reviewed conference papers

[C1]: K. Wojciechowski, E. Gursoy, A. Haffemayer, **S. Kleff**, V. Bonnet, F. Lamiraux, N. Mansard. "Learning-Guided Force-Feedback Model Predictive Control with Obstacle Avoidance for Robotic Deburring". *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[C2]: A. Jordana*, **S. Kleff***, J. Carpentier, N. Mansard, L. Righetti. "Force Feedback Model-Predictive Control via Online Estimation". *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

[C3]: **S. Kleff**, E. Dantec, G. Saurel, N. Mansard, L. Righetti. "Introducing Force Feedback in Model Predictive Control". *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.

[C4]: **S. Kleff**, J. Carpentier, N. Mansard, L. Righetti. "On the Derivation of the Contact Dynamics in Arbitrary Frames (Application to Polishing with Talos)". *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2022.

[C5]: A. Parag, **S. Kleff**, L. Saci, N. Mansard, O. Stasse. "Value Learning from Trajectory Optimization and Sobolev Descent: A Step Toward Reinforcement Learning with Superlinear Convergence Properties". *IEEE International Conference on Robotics and Automation (ICRA)*, 2022.

[C6]: A. Gazar, M. Khadiv, **S. Kleff**, A. Del Prete, L. Righetti. "Nonlinear Stochastic Trajectory Optimization for Centroidal Momentum Motion Generation of Legged Robots". *International Symposium of Robotics Research (ISRR)*, 2022.

[C7]: **S. Kleff**, A. Meduri, R. Budhiraja, N. Mansard, L. Righetti. "High-Frequency Nonlinear Model Predictive Control of a Manipulator". *IEEE International Conference on Robotics and Automation (ICRA)*, 2021. (∼115 citations.)

[C8]: **S. Kleff**, N. Li. "A Sampled-Data Hamilton-Jacobi Reachability Approach to Safe and Robust Motion Planning". *Chinese Control and Decision Conference (CCDC)*, 2018.

[C9]: M. Khadiv, **S. Kleff**, A. Herzog, S. A. A. Moosavian, S. Schaal, L. Righetti. "Stepping Stabilization Using a Combination of DCM Tracking and Step Adjustment". *RSI International Conference on Robotics and Mechatronics (ICRoM)*, 2017.

National conferences (peer-reviewed)

[N1]: **S. Kleff**, E. Dantec. "Contrôle prédictif sur robots commandés en couple: résultats expérimentaux". *Journées Nationales de la Robotique Humanoïde (JNRH)*, 2021.

Manuscripts in preparation

[P1]: E. Gursoy*, K. Wojciechowski*, M. Fourmy, A. Haffemayer, **S. Kleff**, V. Petrik, J. Cavalcanti Santos, V. Bonnet, N. Mansard. "Force-Feedback Model Predictive Control with Diffusion Motion Priors for Contact-Rich Industrial Tasks". In preparation, 2026.

[P2]: P. Puchaud, **S. Kleff**. "Analytical muscle torques derivatives". Under revision for the 51th *Biomechanics Congress*, 2026

[P3]: A. Barsacq, **S. Kleff**, P. Puchaud, J.-C. Sarrazin, D. Daney. "Ghostly Noise: a Sensorimotor Model for the Ouija Experience". Under revision for the 51th *Biomechanics Congress*, 2026.

PhD dissertation

[T1]: **S. Kleff**. *Toward Force Feedback in Model-Predictive Control*. PhD thesis, New York University, 2024.

Selected software

Summary: I develop reusable open-source software tools for nonlinear MPC in robotics, with emphasis on real-time execution and experimental reproducibility. My main contributions span core solver development, NMPC validation pipelines and pedagogical materials for community onboarding.

Main developer

`mim_solvers` *Since Nov. 2023*

C++/Python SQP-based numerical optimal-control solver providing a high-performance back-end for `crocoddyl` [J2]. Enables real-time constrained NMPC on manipulators, quadrupeds, humanoids and drones. Used by 20+ researchers across 4+ groups; supported the EU project AGIMUS.

Main developer

force_feedback_mpc *Since 2022*

First experimentally validated implementation of force-feedback NMPC on modern manipulators [C3], [JR1]. Interfaced with `crocoddyl` and used in the EU project AGIMUS for industrial deburring [C1] and polishing tasks. Companion repository `force_feedback_dgh` provides real-time deployment code for the Kuka iiwa.

Contributor

crocoddyl *Since Jan. 2022*

Widely-used optimal-control library for robotics (1.2k stars, 200+ forks), developed at LAAS-CNRS and Heriot-Watt University. I contributed the analytical derivatives of rigid contact dynamics in moving coordinates [C4] (C++ implementation, Python bindings, tests), now a standard feature enabling contact tasks (manipulation, locomotion) in arbitrary frames.

Main developer

croco_mpc_utils *Since 2023*

Python plugin to `crocoddyl` providing optimal-control problem prototyping utilities from templated YAML configuration files. Wraps Crocoddyl's modular API into a simplified interface, reducing model instantiation overhead. Used across several of my contributions and within the AGIMUS software stack.

Dissemination: I also maintain `mpc_tutorial` and `minimal_examples_crocoddyl`, open-source materials lowering the entry barrier to NMPC for students and researchers. Used by 10+ researchers across labs (Inria, CNRS, NYU).

Research projects

Scientific contributor to EU & national projects:

I contributed to **EU projects CONT-ACT**, **MEMMO**, **AGIMUS** and to the **ANR project COURRIER** through scientific publications, deliverables, and dissemination. I conducted a live NMPC demonstration on the Talos humanoid robot during the final evaluation of MEMMO at LAAS–CNRS (2022). The methods developed during my PhD are actively used in AGIMUS for industrial tasks [C1], in collaboration with Airbus.

Supervision and mentoring

Summary: Formal co-supervision (50%) of 1 PhD student and 4 M.Sc. students; scientific mentoring of 4 PhD students and 2 research engineers. Supervision typically combines weekly meetings, whiteboard derivations, paper reading sessions, hands-on coding, and support during experimental validation.

Formal supervision

PhD student, Inria

Pierre Fabre *Since Aug. 2026*

Contact-implicit MPC. Co-advisor (33%) with Justin Carpentier (Inria WILLOW) and Vincent Padois (Inria AUCTUS).

PhD student, Inria

Tianyi Jin *Since Oct. 2025*

Modular MPC architectures for human–robot collaboration. Co-advisor (50%) with Vincent Padois.

M.Sc. student, Inria

Mohammed Omar *Since Jun. 2026*

Muscle actuation modeling for biomechanics. Co-advisor (50%) with Pierre Puchaud.

M.Sc. student, Inria

Kloé Bonnet *Apr. 2025 – Aug. 2025*

Scenario-based MPC for human–robot collaboration. Co-advisor (50%) with Vincent Padois.
Kloé is now a PhD student at Inria (France).

M.Sc. student, LAAS-CNRS

Lison Kardassevitch *Apr. 2022 – Aug. 2022*

Modeling humanoid self-collision avoidance using learned ego-motion maps. Co-advisor (50%) with Nicolas Mansard.
PhD student at Université de Montréal (Canada).

M.Sc. student, LAAS-CNRS

Hakim Cherfi *Apr. 2021 – Aug. 2021*

Parallelization of differential dynamic programming. Co-advisor (50%) with Ewen Dantec.
Hakim defended his PhD at ENAC (France) in Dec. 2025.

Scientific mentoring

Mentoring activities: Sustained scientific guidance through regular meetings, whiteboard derivations, paper reading sessions, code-level discussions, and on-site research visits. I have become a frequent scientific reference on NMPC for early-career researchers entering the field (R4 seminar, pedagogical repo). In Nov. 2025 I coordinated a 1-week internal bootcamp at Inria AUCTUS (code sprints, tutorials, simulation in MuJoCo, ROS2 integration, hardware deployment with `crocoddyl`/`acados`), training the team on real-robot NMPC workflows.

PhD student, LAAS-CNRS — **Toulouse, France**
Arthur Haffemayer — *2023 – 2025*
Scientific guidance on constrained optimal-control formulations for the EU project AGIMUS. Supervised his software contributions to shared tools (`croco_mpc_utils`, `mim_robots`) during a one-month research visit at NYU (Jul. 2024).

Research engineer, LAAS-CNRS — **Toulouse, France**
Krzysztof Wojciechowski — *Since 2024*
Scientific and technical guidance on NMPC, including hardware experimental deployment of force-feedback MPC originating from my PhD. A 3-day on-site visit at LAAS (Jun. 2025) led to one publication [C1] and one in preparation [P1].

Research engineer, Inria — **Talence, France**
Erwann Landais — *Since 2025*
Scientific guidance on collision-aware optimization for robotic manipulators (Défi Transfert Robotique GRIP4ALL), via bi-weekly working sessions.

Other mentees: Alicia Barsacq (PhD, Inria, 2025–current), Ludovic de Matteis (PhD, LAAS-CNRS, 2023), Maxime Sabbah (PhD, LAAS-CNRS, 2024) – guidance on optimal-control formulations and contact modeling.

Teaching

Teaching assistant — **Toulouse, France**
Université Paul Sabatier III, Faculté de Sciences et Ingénierie (L2 EEA). — *Jan. 2022 – Apr. 2022*
"Systèmes à événements discrets" (Automation)

Teaching assistant — **Toulouse, France**
Université Paul Sabatier III, Faculté de Sciences et Ingénierie (L3 EEA). — *Sep. 2021 – Dec. 2021*
"Informatique Industrielle" (C programming)

Teaching assistant — **Online**
Reinforcement Learning Virtual School (RLVS) — *Mar. 2021*
Organized by ANITI and sponsored by Google DeepMind. Approx. 50h; technical support, answering participants' questions during class, and assisting with the online course infrastructure. 1500 participants from 43 countries.

Communications

Talks at international conferences

International Conference on Robotics and Automation (ICRA) — **Vienna, Austria**
Structure-Exploiting Sequential Quadratic Programming for Model-Predictive Control [J2] — *Jun. 2026*

International Conference on Robotics and Automation (ICRA) — **Yokohama, Japan**
Force-Feedback Model-Predictive Control via Online Estimation [C2] — *May 2024*

International Conference on Humanoid Robots (Humanoids) — **Okinawa, Japan**
On the Derivation of the Contact Dynamics in Arbitrary Frames [C4] — *Nov. 2022*

International Conference on Intelligent Robots and Systems (IROS) — **Kyoto, Japan**
Introducing Force Feedback in Model-Predictive Control [C3] — *Oct. 2022*

International Conference on Robotics and Automation (ICRA) — **Xi'an, China**
High-Frequency Nonlinear Model-Predictive Control of a Manipulator [C7] — *Jun. 2021*

Chinese Control and Decision Conference (CCDC) — **Shenyang, China**
A Sampled-Data HJ Reachability Approach to Safe and Robust Motion Planning [C8] — *Jun. 2018*

Invited talks at universities

WILLOW team, Inria — **Paris, France**
From Model-Predictive Control to Real-World Physical Interaction — *Feb. 2026*

DEFROST team, Centre Inria de l'Université de Lille — **Lille, France**
MPC for Physical Interaction: Force Feedback, Tactile Sensing, and Future Directions — *Dec. 2025*

IDH team, LIRMM-CNRS — **Montpellier, France**
MPC for Physical Interaction: Force Feedback, Tactile Sensing, and Future Directions — *Dec. 2025*

Centre Inria de l'Université de Bordeaux — **Talence, France**
DIGEST Seminar on Quadratic Optimization for Robotics — *Sep. 2025*

Gepetto team, LAAS-CNRS — **Toulouse, France**
Invited talk on Force-Feedback MPC for Physical Interaction — *May 2025*

Robot Intelligence Laboratory, Technical University of Munich (TUM) — **Munich, Germany**
Invited talk on Force-Feedback MPC for Physical Interaction — *Jan. 2025*

Computational Robotics Group, Harvard University — **Cambridge, USA**
Toward Force Feedback in Model-Predictive Control — *Jul. 2024*

Biomimetic Robotics Lab, Massachusetts Institute of Technology (MIT) — **Cambridge, USA**
Toward Force Feedback in Model-Predictive Control — *Jul. 2024*

Robotic Manipulation and Mobility Lab, Columbia University — **New York, USA**
Toward Force Feedback in Model-Predictive Control — *Feb. 2024*

AUCTUS team, Inria — **Bordeaux, France**
Toward Force Feedback in Model-Predictive Control — *Feb. 2024*

Regional robotics network of Nouvelle-Aquitaine (R4) — **Online**
Invited talk on Force-Feedback MPC — *Jan. 2023*

Workshops, colloquiums and local conferences

Journées Nationales de la Robotique Humanoïde (JNRH 2026) — **Montpellier, France**
Toward Structured Physical Interaction in Robotics: from Force to Tactile Feedback [J1] — *Jun. 2026*

ATOMPC Workshop at IEEE ICRA 2024 — **Yokohama, Japan**
Stagewise Sequential Quadratic Programming for Optimal Control (poster) [J2] — *May 2024*

Northeast Robotics Colloquium, Yale University — **New Haven, USA**
Stagewise Sequential Quadratic Programming for Optimal Control (poster) [J2] — *Nov. 2023*

Journées Nationales de la Robotique Humanoïde (JNRH 2022) — **Angers, France**
Introducing Force Feedback in Model-Predictive Control [C3] — *Jul. 2022*

Journées Nationales de la Robotique Humanoïde (JNRH 2021) — **Angers, France**
Predictive Control on Torque-Controlled Robots: Experimental Results [C7] — *Jun. 2021*

Academic service

Associate Editor
IEEE International Conference on Intelligent Robots and Systems (IROS) — *Since 2026*

Reviewer
IEEE Transactions on Robotics (T-RO) — *Since 2024*

Reviewer
IEEE Robotics and Automation Letters (RA-L) and IROS — *Since 2022*

Reviewer
IEEE International Conference on Robotics and Automation (ICRA) — *Since 2021*

Reviewer
IEEE Chinese Conference on Decision and Control (CCDC) — *2018*

Outreach and engagement

Summary: I contribute to public outreach and educational initiatives aimed at making robotics and science accessible to non-specialist audiences, particularly school students and communities outside major research hubs.

I also disseminate robotics through tutorials and engage with broader initiatives on responsible innovation.

Competitively selected participant — **Malmö, Sweden**
Responsible AI for Peace and Security — *Nov. 2023*
Workshop co-organized by the Stockholm International Peace Research Institute (SIPRI) and the United Nations Office for Disarmament Affairs (UNODA). Part of a broader EU-supported initiative on responsible AI innovation, which culminated in the 2025 SIPRI report.

Invited tutorial on optimal control — **Online**
IDESSAI Inria-DFKI Summer School — *Oct. 2025*
Event postponed; course material publicly available.

Outreach workshop — **Lycée Stendhal, Aiguillon, France**
Fête de la Science – Circuit Scientifique — *Oct. 2025*
Hands-on workshop with passive robotic arms, enabling high-school students to manually explore basic robot kinematics.

Robotics outreach for primary-school students — **Bègles & Carbon Blanc, France**
Festival Sciences tout court(s) — *May 2025*

Robotics outreach for primary-school students — **Mérignac, France**
Côté Sciences Air & Espace — *Jan. 2025*

Mentor — **New York University**
NYU ARISE Program — *2020, 2024*
Inclusive STEM education program for high-school students from underrepresented backgrounds. Designed robotics tutorial materials, defined the research project, weekly meetings, and preparation of final oral presentations and posters.

Leadership and collective responsibilities

Coordinator and representative — **New York, USA**
NYU Researchers United (NYURU) — *2022 – 2024*
Coordinator and representative of 400+ research workers (PhDs, postdocs, staff) at NYU Tandon School of Engineering within the NYU-wide researchers' union (4000+ workers). Participated in monthly university-wide strategic meetings; coordinated outreach and communication across campuses; mentored new representatives; contributed to public communication efforts, including interviews with student media (NYU Researchers United; NYU News)